

APPROVAL SHEET

MODEL NAME	CO SENSOR EXCO-1000-2E
PART NUMBER	20203AN05
CUSTOMER NAME	-
CUSTOMER PART NUMBER	-
DATE	2025.10.28
REMARK	R00
SOFTWARE VERSION	-
SOFTWARE CHECKSUM	-



EXSEN Inc.

	Written	Reviewed	Approved
Signature			
Name			
Date			

Customer

	Written	Reviewed	Approved
Signature			
Name			
Date			

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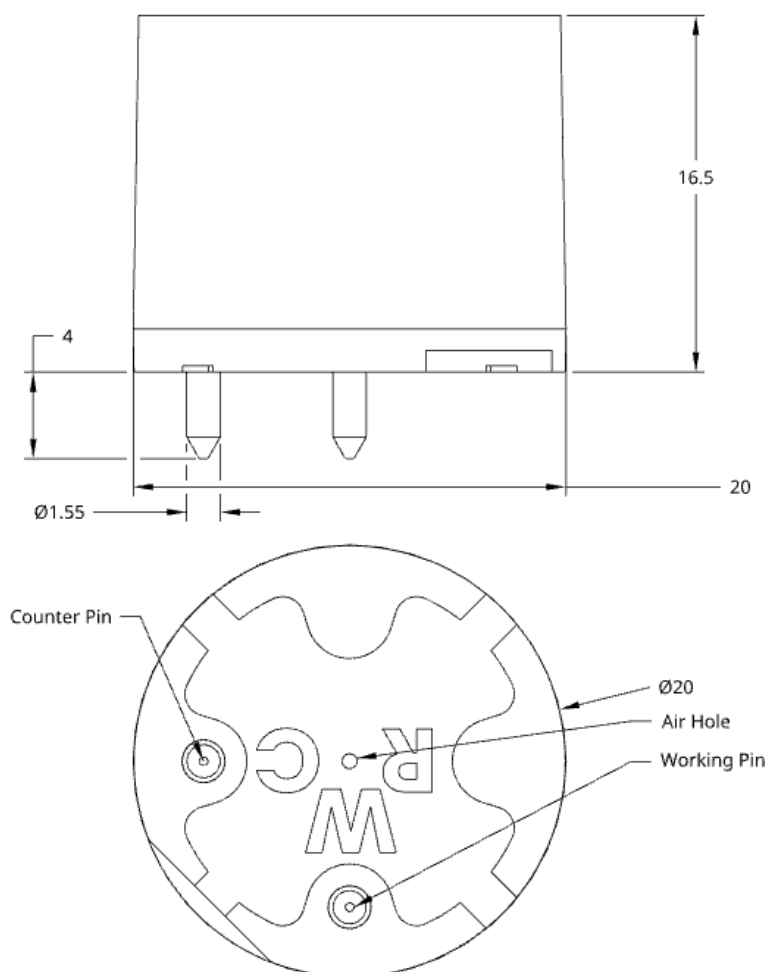
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2. DATA SHEET

MODEL NAME	CO SENSOR EXCO-1000-2E
PART NUMBER	20203AN05
DIMENSION	20Φ x 16.5 mm
DETECTION RANGE	0~500 ppm
COMMUNICATION	Ampere output
APPLICATION	Carbon monoxide concentration display General Purpose

(1) Dimension

- 20Φ x 16.5 (D x H, mm)

**General Tolerance (mm)**

Linear	±0.3
Radius	±0.5

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(2) Sensor & electrical performance specification ($T_a = 25^{\circ}\text{C}$, 50%RH)

Parameters		Condition	Symbol	Min	Typ	Max	Unit
Gas	Target gas	-	T_{Gas}	CO			-
Data	Sensor type	-	EC	2 electrode Electrochemical			
	Detection range	-	DD_R	0-500			ppm
	Maximum range	-	DD_{Max}	1,000			ppm
	Resolution	-	D_R	It depends on OPAMP and receiver's analog input resolution			-
	Baseline Offset	Clean Air	B_O	-2~5			ppm equivalent
	Zero Shift	-10~50 °C	Z_S	< 10			ppm equivalent
	Repeatability	-	R_t	-5	-	5	%
	Linearity	-	L_t	-5	-	5	%
	Sensitivity	At Normal temp, humidity, pressure	D_S	10	15	20	nA/ppm
Time	Response	-	T_{Res}	< 30s for 90% for diffusion sampling method			
	Long term Output Drift	In normal condition	L_{OD}	< 5			%/year
	Storage Life	-	S_L	6 months in manufacturer's packing			-
	Life-time	-	T_{LT}	> 6 years in normal environment, from manufacturing date.			-
Power	Input	Bias is not required	V_{IN}	-	-	-	V
	Current Consumption	-	P_A	-	-	-	A
Electrical	Interface connections	-	O_C	Current output			A
	Sampling interval	-	T_{SPL}	-	Depends on receiver's system		Hz
	Load Resistor	Recommend	L_R	5	10	-	Ohm
	Connector	-	CNT	2 Brass Pin			-
Mechanical	Housing Mat.	-	-	PC(Polycarbonate)			-
	Weight	-	-	5.2			g
	Orientation	-	-	Any			-
Ambient	Operating Temp	-	O_T	-20	25	50	°C
	Operating Humidity	No condensing	O_H	15	-	90	%
	Storage Temp	-	S_T	10	25	30	°C
	Storage Humidity	-	S_H	15	-	90	%
Calibration		-	CAL	Calibration data is written on the sensor package.			-

(3) Interference gas sensitivity

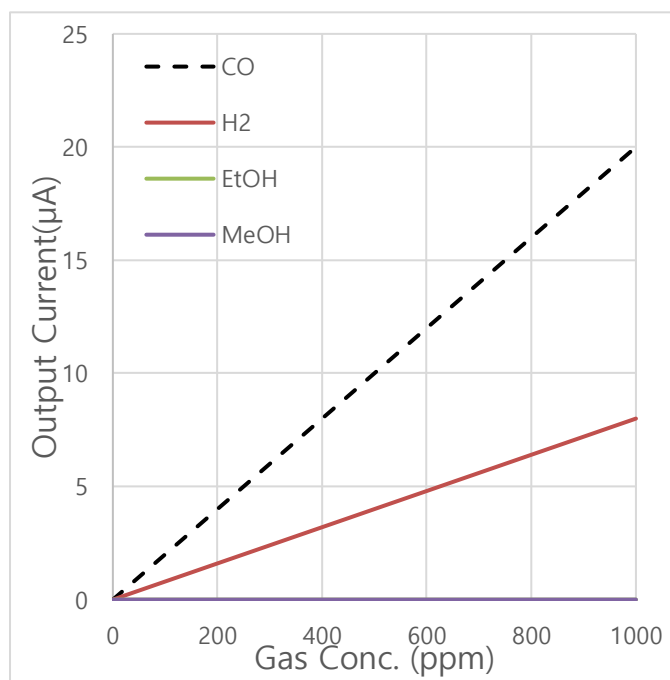


Figure 1 Interference gas sensitivity

Target Gas	Testing Conc.	CO reading value (ppm)	Exposure Time (min)
CO	100	100	30
H ₂	100	40	30
MeOH	1000	0	30
EtOH	1000	<10	30
NH ₃	100	0	5
ETAC	100	0	5
HexaEthylDisiloxane	10	0	40
Toluene	200	0	30
Acetone	200	<1.0	30
Iso-Butane	1000	0	30
CO ₂	1000	0	30
SO ₂	25	0	30
H ₂ S	10	0	30
NO	10	<1.0	5
NO ₂	10	0	30

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(4) Sensor characteristics

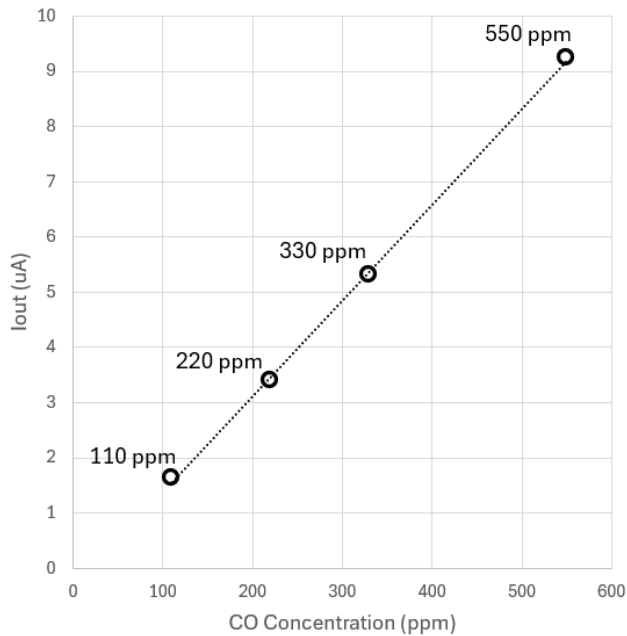


Figure 2 Linearity

EXCO-1000-2E provides highly linear response and reliable accuracy across the 0~500 ppm concentration range.

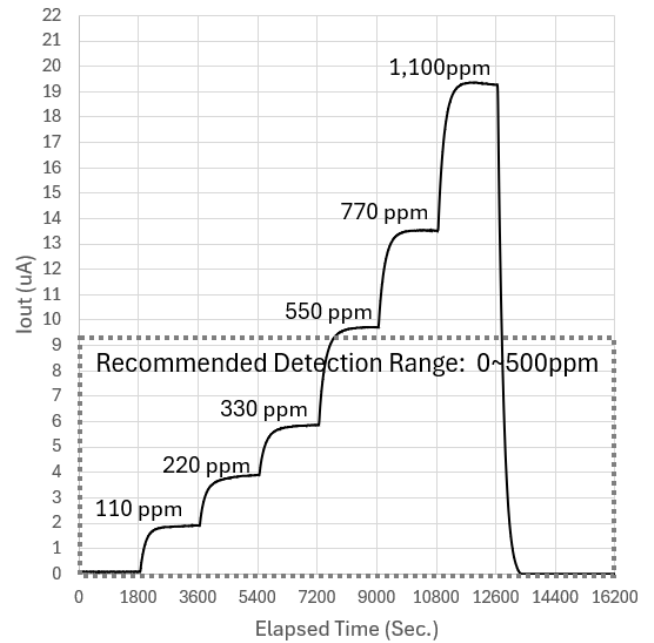


Figure 3 Recommended Detection Range

Measurement Range 0~1000 ppm;

Recommended Operating Range 0~500 ppm

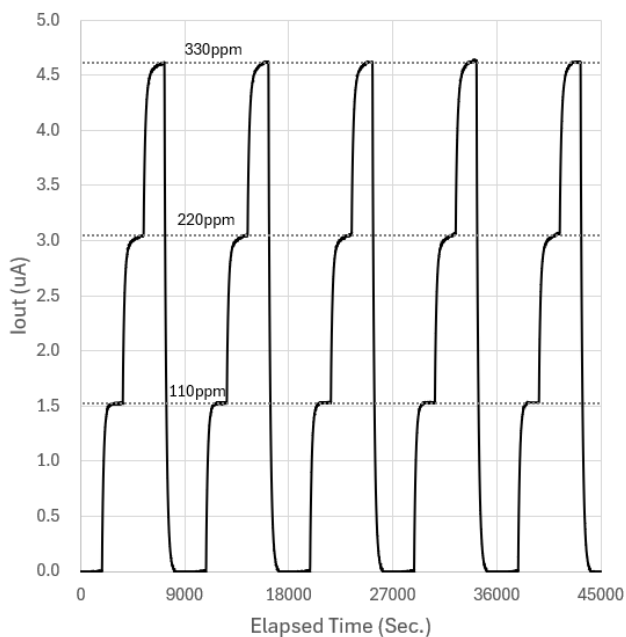


Figure 4 Repeatability

Output response pattern of the sensor under repeated exposure to various CO concentrations at 9000 s intervals, exhibiting outstanding repeatability with less than $\pm 5\%$ deviation

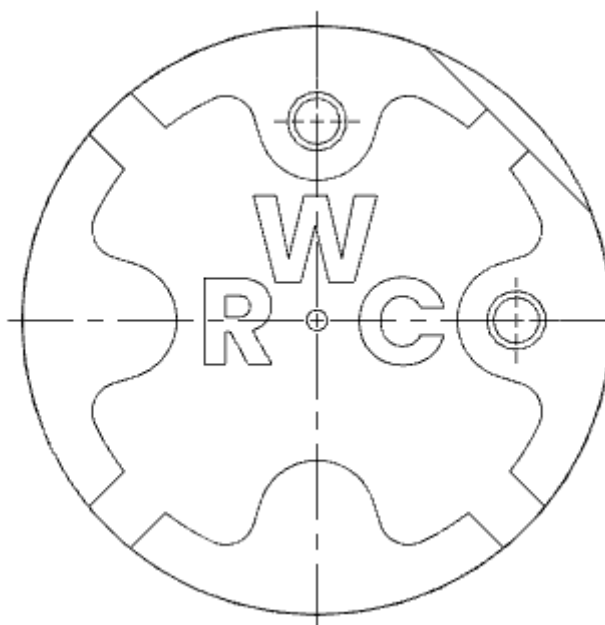
3. Terminal, Label descriptions

- Connector

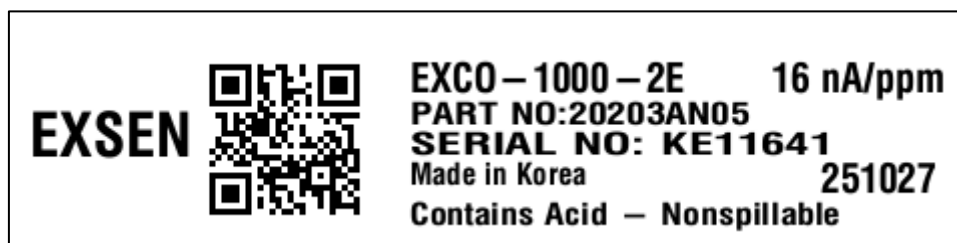
Model name	Type	Pin no
1.55 Diameter Brass Pin	Gold doped Brass Pin	2

Pin No.	Symbol	Description
1	C	Counter Pin
2	W	Working Pin
3	R	Reserved Pin (no Pin)

2. Working Pin



1. Counter Pin



QR Code Data: CO KE1164120203AN0525102716

CO: Target Gas, 2 Characters + 1 blank

KE11641: Serial Number, 8 Characters

20203AN05: Part Number, 9 Characters

251027: Manufacturing Date (YYMMDD), 6 Characters

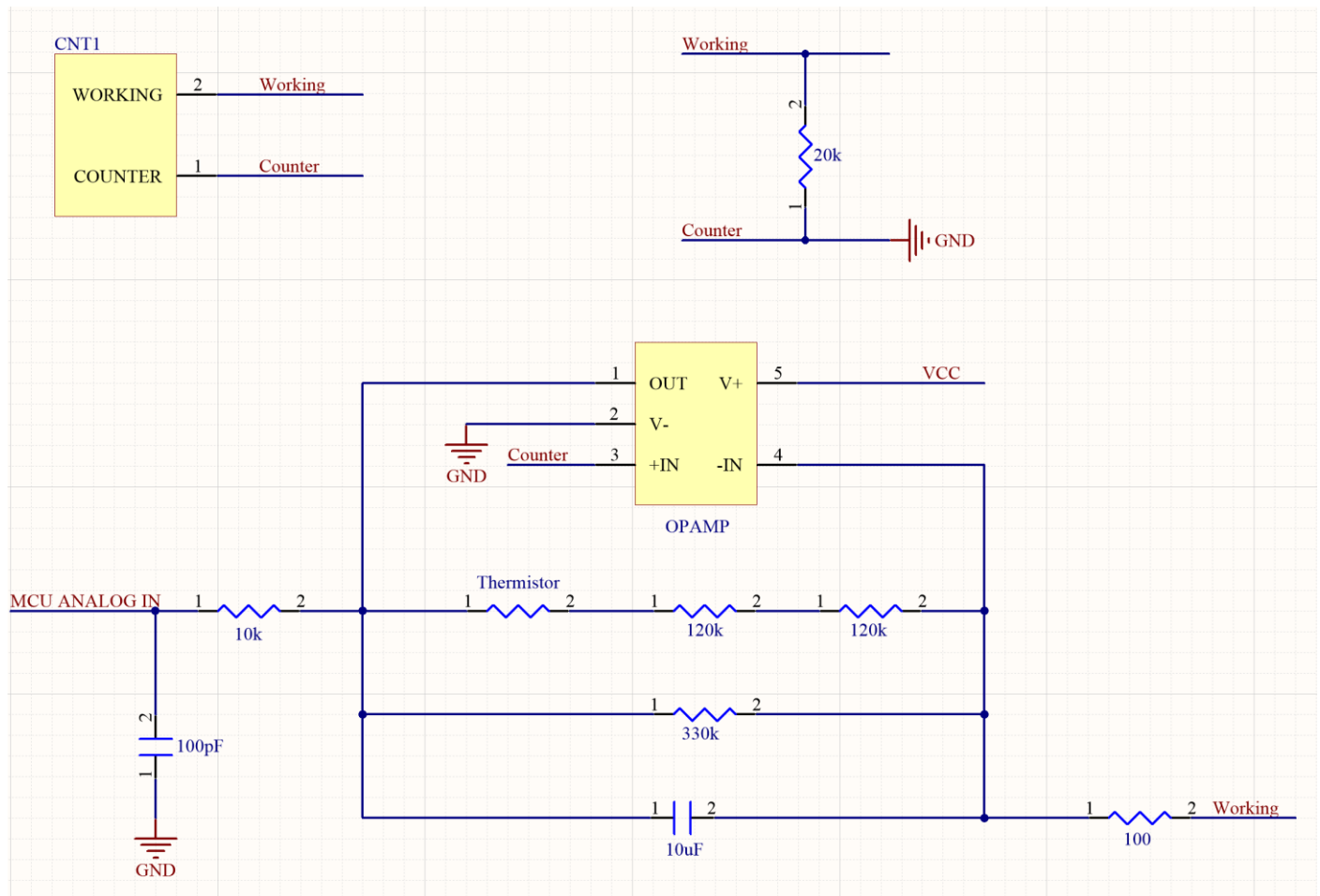
16: Sensitivity (nA/ppm), 2 Characters

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4. Recommended Circuit

(1) Sample Circuit



Recommended Thermistor

R25: 47k

B25/50: 4050

TDK B57356V5473F260 or an equivalent product

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